

# Basic Linux and Git Commands

## 1. Basic Linux Commands

At least 10 commands are provided below, with sample usage and a short description in simple wording.

| Command | Sample Usage           | Description                                                                     |
|---------|------------------------|---------------------------------------------------------------------------------|
| pwd     | pwd                    | Shows the folder you are currently working in.                                  |
| ls      | ls -la                 | Lists files and folders. -l gives details; -a also shows hidden items.          |
| cd      | cd Documents           | Moves you to another directory.                                                 |
| mkdir   | mkdir project          | Creates a new folder.                                                           |
| touch   | touch notes.txt        | Creates a new empty file or updates its timestamp.                              |
| cp      | cp file.txt backup.txt | Copies a file or folder to another location.                                    |
| mv      | mv old.txt new.txt     | Moves a file or renames it.                                                     |
| rm      | rm notes.txt           | Deletes a file. Use carefully because it normally does not go to a recycle bin. |
| cat     | cat notes.txt          | Displays the contents of a text file.                                           |
| clear   | clear                  | Cleans the terminal screen.                                                     |
| echo    | echo "Hello World"     | Prints text in the terminal.                                                    |
| grep    | grep "error" log.txt   | Searches text for a matching word or pattern.                                   |

## 2. Basic Git Commands

Each command includes sample usage, a description written in simple wording, and common options or flags.

| Command     | Sample Usage                                  | Description                                                               | Common Options / Flags                                                                         |
|-------------|-----------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| git init    | git init                                      | Starts Git tracking in the current folder.                                | -q = quiet output                                                                              |
| git clone   | git clone<br>https://github.com/user/repo.git | Downloads a repository and creates a local working copy.                  | -b = clone a specific branch; --depth 1 = shallow clone                                        |
| git status  | git status                                    | Shows changed files, staged files, and the current branch.                | -s = short format; -b = show branch information                                                |
| git add     | git add .                                     | Stages changes so they can be included in the next commit.                | -A = stage all changes; -p = choose changes interactively                                      |
| git commit  | git commit -m "Add login page"                | Saves the staged changes as a new point in the project history.           | -m = commit message; -a = stage tracked modified files and commit                              |
| git log     | git log --oneline                             | Displays previous commits and their history.                              | --oneline = compact view; --graph = show branch structure; -n 5 = show 5 commits               |
| git branch  | git branch feature-login                      | Creates or lists branches used to work on different versions/features.    | -a = show local and remote branches; -d = delete a branch                                      |
| git switch  | git switch feature-login                      | Changes the branch you are currently working on.                          | -c = create and switch to a new branch; - = switch to previous branch                          |
| git pull    | git pull origin main                          | Gets remote changes and integrates them into your current branch.         | --rebase = replay local commits on top of fetched changes; --ff-only = only allow fast-forward |
| git push    | git push -u origin main                       | Uploads local commits to a remote repository such as GitHub.              | -u = set upstream branch; --force = overwrite remote history, use with caution                 |
| git remote  | git remote -v                                 | Shows the remote repositories connected to the local project.             | -v = show remote URLs; add = add a remote                                                      |
| git merge   | git merge feature-login                       | Combines the history of another branch into the current branch.           | --no-ff = create a merge commit; --abort = cancel an unfinished merge                          |
| git diff    | git diff                                      | Shows changes that have not yet been staged.                              | --staged = show staged changes; --stat = show a summary                                        |
| git restore | git restore notes.txt                         | Reverts changes in a file that have not been committed.                   | --staged = unstage a file; --source = restore from another commit/branch                       |
| git stash   | git stash                                     | Temporarily stores uncommitted changes so you can work on something else. | pop = restore and remove stash; list = show saved stashes; apply = restore without removing    |

**Example Git workflow:** git status → git add . → git commit -m "Update project" → git push

**Reminder:** Be careful with commands such as **git push --force** and **rm** because they can cause unwanted changes or data loss.